

Enhanced natural parameterized quantum circuit

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The classical-quantum interface for loading classical data into quantum systems is an indispensable component of quantum information processing, and the parameterized quantum circuit (PQC) represents significant methodology in this area. However, a challenge in designing parameterized quantum circuits that can fully utilize limited qubit resources while accurately representing classical data remains. To address this issue, we propose the enhanced natural parameterized quantum circuit (ENPQC), which achieves the maximum parameter capacity of quantum systems and preserves the local structure of the original dataset. To make the encoding circuits more experiment friendly, we further provide two near-optimal circuits that achieve near-maximum capacity and are comparable in complexity to the existing PQCs. We numerically show that the ENPQC overwhelms other encoding methods for multiple datasets, demonstrating its potential for machine learning tasks. Furthermore, we experimentally validate our scheme by demonstrating a three-class classification task on an NMR platform, which achieves over 97% accuracy. This work provides a powerful tool for the classical-quantum interface, paving the way for quantum big data processing during the noisy intermediate-scale quantum era.

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I. INTRODUCTION

Quantum computing and quantum information provide a tremendous opportunity to revolutionize the way we deal with big data [1,2] due to their exponential acceleration capability and their enhanced data encryption and security mechanisms [3,4]. In order to leverage the superiority of quantum technologies for classical tasks, a crucial prerequisite is determining how to efficiently load classical data into quantum systems [5]. A good classical-quantum interface can efficiently encode more classical data into a quantum device by consuming fewer quantum resources, including fewer qubits and shorter control sequences. Moreover, the encoded quantum states should reflect the topological relationships of the original data, which has a great impact on the functionality and accuracy of subsequent quantum data processing tasks [6,7].

In the early stages, classical data were directly mapped to the state vector of qubits and loaded into a quantum device through quantum state preparation [8–14]. This encoding strategy encompasses various schemes such as basis encoding [7], amplitude/phase encoding [15–18], and quantum random access memory [19]. Since the classical data are directly stored in explicit quantum entities, such as polarizations and phases, the encoding scheme has clear physical meaning [20–22]. However, the biggest drawback is that the preparation of general quantum states with high fidelity is always a challenge task, especially in cases involving a large number of qubits [9,13]. Therefore, the classical-quantum interface based on state preparation gradually becomes inadequate for the task of loading large datasets to quantum devices, particularly as we enter the noisy intermediate-scale quantum (NISQ) era [23–26].

In recent years, heuristic encoding strategies based on parameterized quantum circuits (PQCs) have garnered increasing attention [27,28]. As depicted in Fig. 1(a), by encoding the classical data x into the rotation angles θ of the single-qubit gates, the PQC-based interface can circumvent the requirement for preparing target states and load the dataset with high efficiency [6,29,30]. In general, the encoded states at the end of the PQC will contain the information about x in an abstract and nonlinear manner. A novel encoding scheme known as the natural parameterized quantum circuit (NPQC) has been proposed [31]. Within this framework, the distance measure between the classical input data is maintained in the

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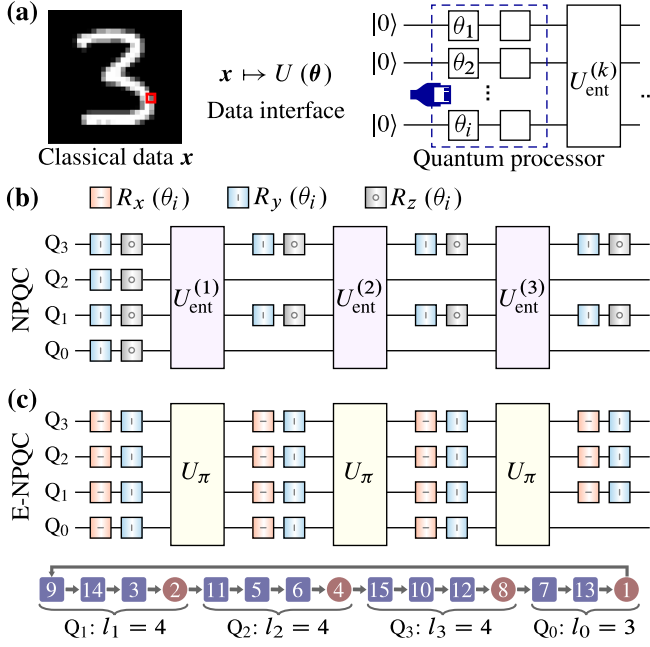


FIG. 1. PQC-based classical-quantum interfaces and the comparison of two NPQC encoding schemes. (a) A schematic diagram illustrating the loading of classical data x into a quantum system using a general PQC. Here, θ_i denotes the rotation angles of the single-qubit gates, and U_{ent} is the entangling gate. (b) An example of the original NPQC encoding scheme, which enables the loading of at most 20 classical parameters into a four-qubit system. (c) Our proposed ENPQC encoding scheme for a four-qubit system. The general permutation U_π is employed as the entangling gate U_{ent} , and its permutation relation is depicted at the bottom. The encoding scheme reaches the maximum parameter capacity (the degree of freedom of the four-qubit pure state).

encoded quantum states. The input data can be transformed into a higher-dimensional space, where they become linearly separable, consistent with the widely utilized Gaussian kernel function in machine learning applications. Therefore, the NPQC serves as a classical-quantum interface with both high encoding efficiency and good interpretability.

However, compared to other encoding schemes, the original NPQCs do not exhibit a clear advantage in parameter capacity, which denotes the amount of data that can be independently loaded and is upper bounded by the degree of freedom (DOF) of the quantum device's pure state [32–36]. Improving the encoding capacity could help us fully utilize the Hilbert space of quantum devices and effectively reduce the width and depth of the encoding circuits [35], which is of great significance for NISQ data processing tasks. For the NPQC scheme, the encoding capacity is closely related to the properties of entangling gates in the circuit [31], but it is still an open question whether and how the maximum encoding capacity can be achieved.

In this work, we demonstrate in both theory and experiment an enhanced classical-quantum interface based on an ansatz that achieves the maximum encoding capacity. First, we theoretically prove that designing special permutations as the entangling gates allows for achieving the maximum encoding capacity of $2^{N+1} - 2$ in N -qubit PQCs. Considering that

the implementation of permutations is not straightforward in practical experiments, we further propose specific entangling structures containing only CNOT gates which can achieve the capacity near maximum value. Based on the nuclear magnetic resonance (NMR) quantum platform, we then experimentally demonstrate our encoding scheme in a three-class classification task and achieve 97% classification accuracy on the test set. Our work develops a PQC-based encoding strategy with high efficiency, good interpretability, and maximum capacity, which provides a superior classical-quantum interface for quantum machine learning and quantum big data processing.

II. ENHANCED NPQC SCHEME

In PQC-based classical-quantum interfaces [29,30], the quantum circuit architecture comprises layers that include single-qubit rotations denoted by (R_x, R_y, R_z) , along with an entangling gate designated as U_{ent} . The classical input data x are mapped to the rotation angles of these qubits through a linear transformation represented by $\theta = \theta_r + cx$; this encoding process is illustrated in Fig. 1(a). Here, x represents a classical feature vector, θ_r represents the reference parameters, and c is a scaling coefficient. In this way, one not only can load the classical information into the qubits but can also define a kernel function that connects the original data space with the Hilbert space of the quantum state. Specifically, under the condition $\|cx\| \ll 1$, the fidelity between two encoded states simplifies into a form resembling a weighted Gaussian kernel function [37], signifying the PQC as the corresponding feature mapping:

$$|\langle \psi(\theta_i) | \psi(\theta_j) \rangle|^2 \approx e^{-\frac{c^2}{4}(x_i - x_j)^T \mathcal{F}(\theta_r)(x_i - x_j)}. \quad (1)$$

Here, $\mathcal{F}(\theta_r)$ is the quantum Fisher information matrix, defined as $\mathcal{F}_{ij}(\theta) = 4\Re[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle]$, with $|\partial_i \psi\rangle = \partial |\psi(\theta)\rangle / \partial \theta_i$ [38,39]. Furthermore, if $\mathcal{F}(\theta_r) = \mathbb{1}$ for a certain circuit, the inner product of two encoded states corresponds to the standard Gaussian kernel that is commonly utilized in various machine learning applications. This variant of PQC, called the natural PQC, has a Euclidean quantum geometry at the reference parameter θ_r , facilitating faster training of variational quantum algorithms and enhancing quantum metrology applications [31].

The number of noncorrelated classical parameters embedded in the quantum state is limited by the DOF of the quantum system. Here, the effective quantum dimension $G_C(\theta) = \text{rank}[\mathcal{F}(\theta)]$ is defined to measure the parameter capacity of a given N -qubit circuit [32]. As for the NPQC structure that maintains the condition $\mathcal{F}(\theta_r) = \mathbb{1}$, we can gradually increase the number of circuit layers until the parameter capacity is saturated. At this point, the maximum value of $G_C(\theta_r)$, denoted as C_N , can be defined as the structure's NPQC capacity. For example, as depicted in Fig. 1(b), the original NPQC architecture tends to be sparse, and in most layers only $N/2$ qubits are utilized. The maximal capacity of the NPQC is $N(2^{N/2} + 1)$ [31], indicating that the number of independent parameters will not increase with the addition of extra layers.

A. Analytical solution for PQC design

Here, we introduce the enhanced NPQC (ENPQC), which aims to maximize the parameter capacity up to the full DOF of the N -qubit pure state ($2^{N+1} - 2$) while maintaining the Euclidean quantum geometry, i.e., $\mathcal{F}(\theta_r) = \mathbb{1}$. As illustrated in Fig. 1(c), the ENPQC consists of single-qubit rotations and an entangling gate per layer, with the encoded state represented as $|\psi(\theta)\rangle = \prod_{l=0}^{L-1} U_l(\theta)|\bar{0}\rangle$, where $U_l(\theta) = U_{\text{ent}} \prod_{n=0}^{N-1} R_x(\theta_{l,x}^{(n)}) R_y(\theta_{l,y}^{(n)})$ and N and L denote the number of qubits and circuit layers, respectively. Throughout this work, we adopt zero-based numbering and use little-endian qubit ordering, with $|\bar{n}\rangle$ ($n = 0, 1, \dots, 2^N - 1$) representing the computational basis states, where $\bar{n}$ is the N -digit binary representation of n .

Consider a scenario where the reference parameters $\theta_r = \mathbf{0}$ and the entanglement gate satisfies $U_{\text{ent}}|\bar{0}\rangle = |\bar{0}\rangle$, a property commonly observed in an ansatz constructed from controlled gates. Thus, the matrix element of $\mathcal{F}(\theta_r)$ associated with the parameters $\theta_{l,\alpha}^{(n)}$ and $\theta_{l',\beta}^{(n')}$ simplifies to

$$\mathcal{F}_{(l',n',\alpha\beta)}(\mathbf{0}) = \text{Re}(\langle \bar{0} | \hat{\sigma}_\alpha^{(n)} (U_{\text{ent}})^{l-l'} \hat{\sigma}_\beta^{(n')} | \bar{0} \rangle). \quad (2)$$

For $l = l'$, the condition $\mathcal{F}_{(l,l,n',\alpha\beta)}(\mathbf{0}) = \delta_{nn'} \delta_{\alpha\beta}$ ensures that the diagonal elements of $\mathcal{F}(\theta_r)$ are equal to 1. For $l \neq l'$, using $\sigma_x^{(n)}|\bar{0}\rangle = |\bar{2}^n\rangle$ and $\sigma_y^{(n)}|\bar{0}\rangle = i|\bar{2}^n\rangle$, the off-diagonal elements with $\alpha \neq \beta$ result in $\mathcal{F}_{(l',n',\alpha\beta)}(\mathbf{0}) = \text{Re}\{\pm i \langle \bar{2}^n | (U_{\text{ent}})^{l-l'} | \bar{2}^{n'} \rangle\}$, which vanish if U_{ent} is additionally restricted to be real in the computational basis. Consequently, the remaining matrix elements to evaluate are of the form

$$\mathcal{F}_{(l \neq l', n', \alpha\alpha)}(\mathbf{0}) = \langle \bar{2}^n | (U_{\text{ent}})^{l-l'} | \bar{2}^{n'} \rangle. \quad (3)$$

If Eq. (3) yields a nonzero result, it indicates interdependence between the two parameters involved, leading to a mixing of the information they encode; thus, one of the parameters must be removed. This process facilitates the establishment of the layout of parameterized gates in circuits.

Moving forward, we demonstrate in theory that by utilizing the quantum state permutation as the entangling gate ($U_{\text{ent}} = U_\pi$) with a specific design, the parameter capacity of the ENPQC can achieve the DOF of pure states. The details are as follows. The operator U_π acts on the 2^N computational basis states by applying a permutation π , such that $U_\pi|\bar{n}\rangle = |\bar{\pi(n)}\rangle$. Substituting this into Eq. (3) results in

$$\mathcal{F}_{(l \neq l', n', \alpha\alpha)}(\mathbf{0}) = \mathcal{I}[\pi^{l'-l}(2^n) = 2^{n'}], \quad (4)$$

where $\mathcal{I}[X]$ denotes the indicator function, which equals 1 if X is true and 0 otherwise. Since the permutation can be expressed as a product of disjoint cyclic permutations (or cycles), we divide π into three parts:

$$\pi = (0) \circ [\sigma_1 \cdots \sigma_i \cdots] \circ [\epsilon_1 \cdots \epsilon_i \cdots], \quad (5)$$

where the cycle (0) arises from the condition $U_{\text{ent}}|\bar{0}\rangle = |\bar{0}\rangle$, σ_i represents cycles containing at least one element that is a power of 2, and ϵ_i represents cycles that exclude powers of 2. We define $\{l_n\}_{n=0}^{N-1}$ as a structural characteristic of the permutation π , with its definition and connection to the circuit layout formalized in Proposition 1.

Proposition 1. For a 2^N -permutation π with $\pi(0) = 0$ and l_n ($n = 0, 1, \dots, N-1$) as the minimum positive integer

such that $\pi^{-l_n}(2^n)$ is a power of 2, (a) $\forall 0 \leq n, n' < N$, $0 \leq l < l_n$, $0 \leq l' < l_{n'}$, and $\pi^{l'-l}(2^n) = 2^{n'}$ if and only if $l = l'$ and $n = n'$, and (b) $\forall 0 \leq n < N$, $l \geq l_n$, $\exists 0 \leq n' < N$, $0 \leq l' < l_{n'}$, such that $\pi^{l'-l}(2^n) = 2^{n'}$.

Proof. We consider the cycles containing 2^n and $2^{n'}$. Denote the L cycle that includes 2^n as σ_p .

For part (a), if $2^{n'}$ is not in σ_p , then clearly, $\pi^{l'-l}(2^n) \neq 2^{n'}$. If $2^{n'}$ is in σ_p , we consider two cases:

(1) When $n = n'$, the equation $\sigma_p^{l'-l}(2^n) = 2^n$ holds if and only if $l' - l = kL$ for some integer k . It holds only when $l = l'$ since $L \geq l_n$ and $0 \leq l, l' < l_n$.

(2) When $n \neq n'$, the definition of l_n implies that $\sigma_p^K(2^n) = 2^{n'}$ requires $K \geq l_{n'}$ or $K \leq -l_{n'}$, shown as follows:

$$\sigma_p = (\underbrace{\square \square \cdots \square \square}_{\geq l_n} 2^n \underbrace{\square \cdots \square \square}_{\geq l_{n'}} 2^{n'}).$$

Given $0 \leq l < l_n$ and $0 \leq l' < l_{n'}$, it follows that $-l_n < l' - l < l_{n'}$, so $\sigma_p^{l'-l}(2^n) \neq 2^{n'}$.

For part (b), we consider two cases:

(1) If σ_p contains only one power of 2, namely, 2^n , then $L = l_n$. There exist $n' = n$ and $l' = l \bmod L$ such that $\sigma_p^{l'-l}(2^n) = \sigma_p^{kL}(2^n) = 2^n$, $k \in \mathbb{Z}$.

(2) If σ_p contains M powers of 2 where $M \geq 2$, we denote them as $2^{n_0}, 2^{n_1}, \dots, 2^{n_{M-1}}$ in the reverse order of appearance in σ_p , starting from $2^{n_0} = 2^n$,

$$\sigma_p = (\underbrace{\square \cdots \square}_{l_{n_{M-1}}} 2^{n_{M-1}} \cdots \underbrace{\square \cdots \square}_{l_{n_1}} 2^{n_1} \underbrace{\square \cdots \square}_{l_{n_0}} 2^{n_0}).$$

Thus, σ_p is divided into M disjoint intervals, and $\sigma_p^{-l}(2^n)$ must be in one of them. Formally, we define

$$L_j = \begin{cases} 0 & j = 0, \\ \sum_{i=0}^{j-1} l_{n_i} & 1 \leq j < M, \\ L & j = M. \end{cases}$$

For $l \geq l_n$, there exists an integer $0 \leq q < M$ such that $kL + L_q \leq l < kL + L_{q+1}$ for some $k \in \mathbb{N}$. Consequently, there exist $n' = n_q$ and $l' = (l - L_q) \bmod L$ such that $\sigma_p^{l'-l}(2^n) = 2^{n'}$. ■

Proposition 1 implies that if the parameters $\theta_{l,\alpha}^{(n)}$ with $l < l_n$ are chosen to be independent for all n , then all other parameters become redundant in terms of those. Consequently, the NPQC capacity C_N depends on the sum of cycle lengths having powers of 2: $C_N = 2 \sum_{n=0}^{N-1} l_n$. The maximum value of C_N , which equals the DOF of N -qubit pure states $2(2^N - 1)$, is attained if all cycles in π except (0) contain powers of 2. This implies that all ϵ_i terms within Eq. (5) are empty. Therefore, we have found a special class of entanglement gates U_π for NPQC and the corresponding circuit construction scheme. Our ENPQC ansatz is given based on the following rules:

(1) Each layer of the ENPQC ansatz consists of single-qubit rotations $\{R_x, R_y\}$ and an entangling gate U_π , with at most one set of $\{R_x, R_y\}$ on each qubit per layer.

(2) There are l_n layers of $\{R_x, R_y\}$ on the n th qubit Q_n .

When designing a circuit, we can first construct the permutation π , thereby obtaining $\{l_n\}_{n=0}^{N-1}$, and then obtain the corresponding circuit based on the above two rules. The number of layers is clearly bounded by $\max_{n=0}^{N-1} l_n^{N-1}$, with a minimum of $\lceil (2^N - 1)/N \rceil$ when C_N equals the DOF,

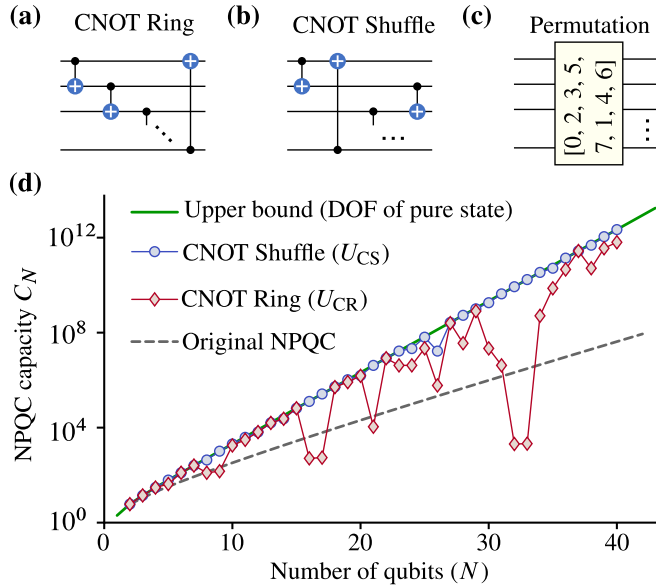


FIG. 2. Different types of entangling patterns and a comparison of their NPQC capacities. The entangling patterns (a) CNOT ring U_{CR} and (b) CNOT shuffle U_{CS} and (c) the general permutation U_{π} serve as U_{ent} in Fig. 1(a) to form ENPQC. (d) Comparison of the N -qubit NPQC capacity in ENPQC with the optimal solution U_{π} (green solid line, reaching pure-state degrees of freedom), CNOT ring (red scattering), CNOT shuffle (blue scattering), and the original NPQC (gray dashed line).

indicating that parameter gates are uniformly distributed across all qubits. In Fig. 1(c), we provide an example of a permutation that achieves the DOF while minimizing the number of circuit layers, with $\{l_n\} (n = 0, 1, 2, 3)$ marked at the bottom. Based on the permutation, we are able to derive the corresponding quantum circuit, as shown in Fig. 1(c).

In Fig. 2(d), the NPQC capacity for various structures is depicted in relation to the number of qubits N , ranging from 2 to 40. The green line signifies the maximum capacity of our ENPQC, equivalent to the degree of freedom of N -qubit pure states, and the dashed line represents parameter capacities of the NPQC scheme raised in previous study.

B. Experiment-friendly near-optimal circuits

The implementation of general permutations in experiment is sometimes challenging [40]; hence, finding some special entangling structures that are experimentally feasible is important. One common strategy is to construct the entangling gates using CNOT gates. Here, we first consider the CNOT ring U_{CR} structure. It involves a consecutive application of CNOT gates among N adjacent qubit pairs when the last qubit is treated as a neighbor to the first one, as shown in Fig. 2(a). We identify a distinctive property of U_{CR} that enables the efficient computation of l_n , as stated in Proposition 2, with proofs provided in the Supplemental Material [41].

Proposition 2. For $1 \leq n < N$, if there exist integers $K > 0$ and $0 \leq m < N$ such that

$$U_{CR}^K |2^n\rangle = |2^m\rangle, \quad (6)$$

then

$$U_{CR}^K |2^{n-1}\rangle = |2^{m-1}\rangle, \quad m \geq 1, \quad (7)$$

$$U_{CR}^{K+1} |2^{n-1}\rangle = |2^{m-1}\rangle, \quad m = 0, \quad (8)$$

and vice versa.

Building on this, it suffices to determine the smallest positive integer K_0 such that $\pi_{CR}^{K_0}(2^{N-1})$ yields a power of 2, denoted by 2^{m_0} ; thus, $l_{m_0} = K_0$. Assuming that $m_0 \geq 1$ without loss of generality, we can deduce from Proposition 2 that $U_{CR}^{K_0} |2^{N-2}\rangle = |2^{m_0-1}\rangle$ and $l_{m_0-1} = K_0$. By analogy, we have

$$l_n = \begin{cases} K_0 & n \leq m_0, \\ K_0 + 1 & n > m_0. \end{cases} \quad (9)$$

Hence, the NPQC capacity $C_N = 2[N(K_0 + 1) - m_0 - 1]$. The encoding part in Fig. 3(a) depicts the three-qubit PQC utilizing U_{CR} , with $K_0 = 2$ and $m_0 = 1$. To determine K_0 and m_0 , we propose two numerical approaches. The first involves computing the matrix representation of U_{CR} to derive the permutation π_{CR} , which incurs exponential computational complexity. The second approach simulates the evolution $U_{CR}^{K_0} |2^{N-1}\rangle = |2^{m_0}\rangle$, where each application of U_{CR} is expressed as a matrix-vector multiplication,

$$\begin{bmatrix} Q'_{N-1} \\ Q'_{N-2} \\ Q'_{N-3} \\ \vdots \\ Q'_0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} Q_{N-1} \\ Q_{N-2} \\ Q_{N-3} \\ \vdots \\ Q_0 \end{bmatrix} \pmod{2}, \quad (10)$$

which reduces the scale from 2^N to N . This makes it possible to handle cases with a larger number of qubits N . In Fig. 2(b), the red diamonds represent the NPQC capacity for the U_{CR} scenario, surpassing the original NPQC in most cases and approaching the pure-state DOF in some instances.

To further improve the parameter capacity, we rearrange the order of CNOT gates in the CNOT ring and construct the entangling gate referred to as CNOT shuffle U_{CS} , as depicted in Fig. 2(b). To construct the associated mapping matrix of U_{CS} , the process begins with an identity matrix, adding the j th row to the k th row modulo 2 for each CNOT(Q_j, Q_k) applied, where row j corresponds to qubit Q_j . As Proposition 2 is inapplicable in universal cases, each l_n ($n = 0, \dots, N-1$) must be computed individually. Given the infeasibility of exhaustively traversing all $N!$ permutations for large N , hundreds of permutations are randomly sampled using Monte Carlo methods, and the one yielding the largest C_N is selected as the estimate for the optimal N -qubit U_{CS} . The results, depicted as blue dots in Fig. 2(d), indicate that the DOF is achieved in most cases simply by rearranging the elements in the CNOT ring. The corresponding configurations are provided in Supplementary Information [41].

The CNOT ring, due to its structure, requires that the CNOT gates be executed sequentially. In contrast, the CNOT shuffle may allow for parallel execution of certain quantum gates depending on the arrangement of the CNOT gates. Notably, the optimal permutation for U_{CS} may not be unique, offering additional flexibility for experimental implementation.

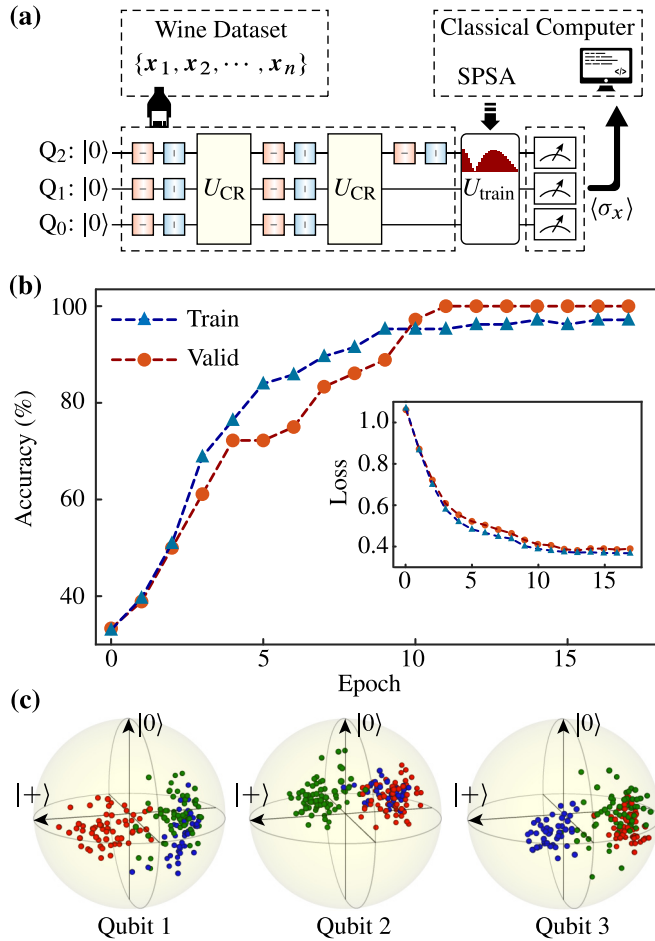


FIG. 3. Experimental demonstration of multiclass classification tasks based on the ENPQC encoding scheme. (a) Schematic diagram of the overall experimental process. The classical wine dataset is divided into training, validation, and test sets in a 6:2:2 ratio. In the three-qubit NMR quantum system, we utilize the ENPQC scheme to encode classical data into quantum states and then determine the final state after applying an operation with adjustable parameters denoted as U_{train} . By optimizing the parameters in U_{train} , we achieve three-class classification of the input data. (b) Iteration curves during the training process. The two lines depict the accuracy of the classification for the training set and validation set at different epochs. (c) Upon completion of training, the simulated reduced density matrices for different qubits in the final states are displayed on the Bloch sphere. Red, green, and blue correspond to the three classes of the input data.

III. EXPERIMENTAL DEMONSTRATION OF THE ENPQC

To demonstrate the experimental feasibility of the ENPQC scheme, we employ this encoding method to perform a three-class classification task on an NMR quantum system. The classical dataset we use is the wine dataset [42,43], comprising 178 instances with chemical analyses for three wine varieties, with each instance containing 13 distinct chemical attributes. Our task involves classifying wine categories based on these attributes. The dataset is split into training, validation, and test sets in a 60:20:20 ratio. The NMR sample used in the experiment is the iodotrifluoroethylene ($\text{C}_2\text{F}_3\text{I}$) dissolved in D-chloroform [44,45], and we utilize three ^{19}F nuclear spins

as a three-qubit quantum processor. The Hamiltonian of the NMR sample is

$$H_s = \sum_i \pi v_i \hat{\sigma}_z^{(i)} + \sum_{ij} \frac{\pi}{2} J_{ij} \hat{\sigma}_z^{(i)} \hat{\sigma}_z^{(j)}, \quad (11)$$

where the parameter v_i denotes the chemical shift associated with the i th spin and J_{ij} indicates the strength of the scalar coupling between the i th and j th spins. Our experiments were performed on a Bruker Avance III 400 MHz spectrometer at room temperature. The system, initially in the thermal equilibrium state, is first prepared as a pseudopure state (PPS) $\hat{\rho}_{\text{pps}} = [(1 - \epsilon)/8]\mathbb{1} + \epsilon|000\rangle\langle 000|$ by using the selective-transition approach [46] with a polarization of $\epsilon \approx 10^{-5}$ prior to loading classical data. The experimental fidelity of $\hat{\rho}_{\text{pps}}$ is about 99.29%.

The experimental process is illustrated in Fig. 3(a). We utilize the CNOT ring as the entangling gate and employ ENPQC to encode classical data $\{x_1, x_2, \dots, x_n\}$ into quantum operations acting on $\hat{\rho}_{\text{pps}}$, resulting in encoded quantum states. For an instance x from the wine dataset, it is transformed by $\theta = \theta_r + cx$ and loaded into the ENPQC to obtain the encoded state $|\psi(\theta)\rangle$. Next, we apply a unitary transformation U_{train} , which is implemented by an NMR shaped pulse sequence consisting of 2000 adjustable control fields $\mathbf{B}_i = (B_x^{(i)}, B_y^{(i)})$ over a duration of $\delta t = 20$ ms. The control Hamiltonian is

$$H_c(\mathbf{B}_i) = \pi \sum_j (B_x^{(i)} \hat{\sigma}_x^{(j)} + B_y^{(i)} \hat{\sigma}_y^{(j)}), \quad (12)$$

and the total evolution operator of the system corresponding to a single shaped pulse is given by $U_i(\mathbf{B}_i) = \exp\{-i[H_s + H_c(\mathbf{B}_i)]\delta t\}$. Thus, $U_{\text{train}} = \prod_i U_i(\mathbf{B}_i)$. The encoded states evolve under U_{train} to the final states, where we measure each qubit's expectation value $\langle\sigma_x\rangle$. The classification of the input data is determined by the maximum measurement value of $\langle\sigma_x\rangle$ across the qubits. Specifically, let the classification result given by the circuit for an input instance be k ($k \in [0, 1, 2]$); we define k as follows:

$$k = \arg \max_i \{ \langle\sigma_x^{(i)}\rangle \}_{i=0,1,2}. \quad (13)$$

Then we define the cost function as follows:

$$\text{Cost}(x_1, \dots, x_n) = \frac{1}{n} \sum_{j=1}^n \left[\frac{1}{3} \sum_{i=0}^2 (y_j^{(i)} - \langle\sigma_x^{(i)}\rangle_j)^2 \right], \quad (14)$$

where i refers to the i th qubit, j indicates the j th instance from the wine dataset, and $y_j^{(i)}$ is the class label for the i th qubit of the j th instance. Minimizing this cost function optimizes U_{train} . We train the parameters $\mathbf{B}$ within U_{train} using the simultaneous perturbation stochastic approximation (SPSA) algorithm [47]. SPSA is an efficient optimization technique that estimates gradients with two measurements per iteration, independent of the parameter count, offering robustness against noise and applicability in black-box scenarios without the need for analytical gradients. Based on the measured expectation values $\langle\sigma_x\rangle$ of each qubit, we calculate the cost function using a classical computer and iterate the parameters with SPSA. Training is conducted with a minibatch size of 4 over 18 epochs [41,48,49]. After the cost function converges, we obtain the trained

parameters $\mathbf{B}$. Using these trained parameters, we can then determine the accuracy of the circuit on the test set.

Figure 3(b) shows the relationship between the classification accuracy of the training set and the validation set, as well as the corresponding loss with iterations of the epoch during the process of training. The final obtained classification accuracies are 97.17% for the training set and 100% for the validation and test sets. Upon completion of training, the reduced density matrix of each qubit in its final state is displayed on the Bloch sphere in Fig. 3(c). Red, green, and blue represent the three classes of input data. In this experiment, the role of U_{train} can be considered a “global rotation” operation that preserves the inner product between any two encoded quantum states invariant. This implies that the distance between final states and U_{train} is unrelated. Consequently, the distances between final states are determined by the encoding scheme used to map classical data into quantum states. To achieve classification tasks, it is essential to distinguish different classes of classical input data as effectively as possible through their corresponding quantum states. If the encoded quantum states are particularly close together, it becomes challenging to find observables suitable for distinguishing them. This means that the encoding schemes directly influence the upper bound on classification accuracy. From Fig. 3(c), it can be observed that our ENPQC scheme effectively maps different categories of data to distinct regions in the state space.

IV. CONCLUSION AND DISCUSSION

In conclusion, we demonstrated in both theory and experiment an efficient classical-quantum interface by establishing a PQC-based encoding framework termed enhanced natural PQC. This encoding scheme maximizes the circuit’s parameter capacity to the degree of freedom of the pure state and preserves the local structure of the original dataset. Our work makes significant progress in addressing the crucial challenge of efficient classical-quantum data transfer during the NISQ era and opens up new possibilities for efficient quantum information processing and quantum machine learning.

To validate the effectiveness of our encoding method, we compared ENPQC with other encoding schemes for quantum classification tasks. As shown by the numerical results in Table I, the ENPQC scheme has significant advantages over the previous amplitude encoding [16,50–52] and angle encoding [18,53,54] methods in terms of qubit number or encoding complexity. Furthermore, in pattern recognition tasks such as the iris [55], wine [43], and breast cancer [56] datasets, the quantum classification using the ENPQC encoding scheme achieves obviously better accuracy. However, for image recognition tasks like handwritten digits [57] and the Fashion-MNIST [58] dataset, the ENPQC scheme performs on par with amplitude encoding and significantly outperforms angle encoding [41]. Overall, the ENPQC encoding scheme demonstrates clear superiority in terms of classification accuracy and resource consumption across various tasks [41].

Our scheme requires the use of permutation operations U_{π} as the entangling gates. Currently, how to efficiently

TABLE I. Comparison of quantum resource consumption and classification accuracy (in percentage) of multiclass classification tasks when using ENPQC with the full DOF, amplitude encoding, and angle encoding. Here, M is the dimension of the classical data. The simulation results for quantum classification tasks for various datasets, including iris, wine, breast cancer, handwritten digits, and Fashion-MNIST, are presented. The training and test sets are split in an 8:2 ratio. The classical data are encoded into quantum states using different methods and then classified using the support vector machine (SVM).

	ENPQC	Amplitude	Angle
Number of qubits	$\mathcal{O}(\log M)$	$\mathcal{O}(\log M)$	$\mathcal{O}(M)$
Iris (%)	97.33	68.00	96.67
Wine (%)	94.41	88.21	92.71
Breast cancer (%)	94.73	80.67	92.61
Handwritten digits (%)	93.93	94.27	48.69
Fashion-MNIST (%)	84.50	84.92	11.39

implement specified permutation operations in a quantum system remains an open question. In Soeken *et al.*’s work [59], a compilation algorithm for quantum state permutations into elementary quantum gates was introduced. Compiling a specific permutation may result in an exponential gate complexity [40]. It is worth noting that the ENPQC ansatz design is related to only $\{l_k\}_{k=0}^{N-1}$, which depend on the relative positions of powers of 2 within the permutation cycles, regardless of other elements. Therefore, we can regard $\{l_k\}$ as the structural feature of the permutation π . This implies that different permutations with the same $\{l_k\}$ have equal impact in terms of the design of encoding the circuit. Therefore, when designing the ENPQC with maximum parameter capacity, we can first construct $\{l_k\}$ such that all ϵ_i terms within Eq. (5) are empty, then obtain different permutations π corresponding to this $\{l_k\}$. The specific choice of which π to use will depend on the physical platform employed. This flexibility facilitates the implementation of ENPQC in various quantum systems.

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DATA AVAILABILITY

Data generated and analyzed during the current study are available from the corresponding author upon reasonable request. All code used in this study is available from the corresponding author upon request.

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